$$P\Big(\Big|\underbrace{\tfrac{1}{n}\sum_{i=1}^{n} X_i}\; - \underbrace{\mu}\Big| > \epsilon\Big) \leq 2\exp(-c n\epsilon^2)$$

$$P\Big(\Big|\underbrace{f(X_1\cdots X_n)}_{Y_n} - \underbrace{\mathbb{E}(f(X_1\cdots X_n))}_{Y_0}\Big| > \epsilon\Big) \leq \underbrace{c_1\exp(-c_2 n\epsilon^2)}_{\to 0}$$

$n \to \infty$

Under what conditions on f this happens

Ch. 2.2 Martingale Based Method

R.V.

$$Y_k := \mathbb{E}\big(\underbrace{f(X_1\cdots X_n)}\,\big|\,\underbrace{X_1\cdots X_k}\big)$$

$\mathbb{E}(X\,|\,Z) \sim$ Assuming that Z is observed, what would be the avg. of X.

$$= \int_{-\infty}^{+\infty} x \cdot \underbrace{f_{X|Z}(x/z)}\,dx = h(Z)$$

scalar

$$Y_0 = \mathbb{E}(f(X_1\cdots X_n)) \sim \text{deterministic}$$

$$Y_n = \mathbb{E}(f(X_1\cdots X_n)\,|\,X_1\cdots X_n) = f(X_1\cdots X_n)$$

$$\underbrace{Y_n - Y_0} = \underbrace{(Y_n - \cancel{Y_{n-1}})}_{D_n} + \underbrace{(\cancel{Y_{n-1}} - \cancel{Y_{n-2}})}_{D_{n-1}} + \cancel{\cdots} + \underbrace{(Y_1 - Y_0)}_{D_1}$$

$$D_k = \underbrace{Y_k - Y_{k-1}} = \mathbb{E}(f\,|\,X_1\cdots X_k) - \mathbb{E}(f\,|\,X_1\cdots X_{k-1})$$

↳ How much of the deviation in $Y_n - Y_0$ is attributed to X_k.

$$\mathbb{E}(D_k\,|\,X_1\cdots X_{k-1}) = \mathbb{E}\Big(\overbrace{\mathbb{E}(f\,|\,X_1\cdots X_k)}^{g(X_1\cdots X_k)} - \underbrace{\mathbb{E}(f\,|\,X_1\cdots X_{k-1})}_{h(X_1\cdots X_{k-1})}\,\Big|\,X_1\cdots X_{k-1}\Big)$$

$$= \int \underbrace{g(x_1\cdots x_k)}\cdot P(x_k\,|\,x_1\cdots x_{k-1})\,dx_k - \mathbb{E}(f\,|\,X_1\cdots X_{k-1})$$

$$\int f(x_1\cdots x_n)\cdot P(x_{k+1}\cdots x_n\,|\,x_1\cdots x_k)\,dx_{k+1}\cdots dx_n \to P(B,C/A)$$

$$= \int f(x_1\cdots x_n)\cdot P(\overbrace{x_{k+1}\cdots x_n}^{C}\,|\,\overbrace{x_1\cdots x_k}^{A,B})\cdot P(\overbrace{x_k}^{B}/\overbrace{x_1\cdots x_{k-1}}^{A})\,dx_k\cdots dx_n$$

$= \int f(x_1 \dots x_n)\, p(x_k, x_{k+1} \dots x_n \mid \underbrace{x_1 \dots x_{k-1}})\, dx_k\, dx_{k+1} \dots dx_n$

$= \mathbb{E}(f(X_1 \dots X_n) \mid X_1 \dots X_{k-1})$

Tower Property of Cond. Expec.

σ- algebra

Prob. Space (Ω, F, P)

Ω → Outcome

F → possible event

P → prob. measure

$F = 2^{\Omega}$

F is a σ-alg. over Ω.

$F \subseteq 2^{\Omega}$

$\Omega \in F$

$A \in F \Rightarrow \bar{A} := \Omega \setminus A \in F$

$A_1 \dots \in F \Rightarrow A_1 \cup A_2 \cup \dots \in F$

Example:

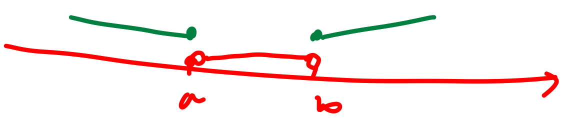
$\Omega = \{1, 2, \dots 6\}$

$\mathcal{F} = \{\{1, \dots, 6\}, \{1,2,3\}, \{4,5,6\}, \{\}\}$ is σ- alg.

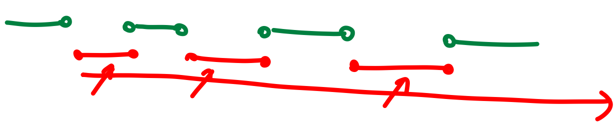
$\mathcal{F} = \{\{1, \dots, 6\}, \{1,2\}, \{3,4\}, \{5,6\}, \{\}\}$ is not σ- alg.

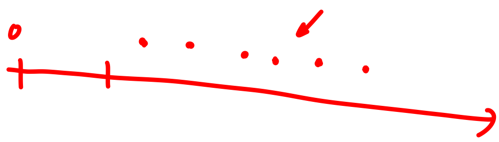
$\Omega = \mathbb{R}$

$\mathcal{F} = \{$ the set of all open intervals $\}$



$\mathcal{F} = \{$ countable union of intervals $\}$ σ-alg.





measurable func.

Can we define any function g on $\mathcal{F}$ such that the output of g could be measured?

$g : \mathcal{F} \to \mathcal{G}$

$\forall e \in \mathcal{G} : \quad g^{-1}(e) \in \mathcal{F}$

Filtration $\quad \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \dots \quad \mathcal{F}_k \subseteq \dots$

$\mathcal{F}_n := \sigma(X_1 \dots X_n) \rightsquigarrow$ all events defined on joint of $X_1 \dots X_n$

$\mathcal{F}_n \subseteq \mathcal{F}_{n+1}$

Def. $\quad Y_1 \dots Y_k$ is adapted to a filtration if Y_k is measurable func. of $(X_1 \dots X_k)$.

Def. $\quad Y_1 \dots Y_k$ are adapted to the filtration $\mathcal{F}_1 \dots \mathcal{F}_k$. Then $(Y_k, \mathcal{F}_k)_{k=1}^{\infty}$ is a martingale

$$\mathbb{E}(|Y_k|) < \infty.$$

$$\mathbb{E}(Y_{k+1} / \underbrace{\mathcal{F}_k}_{= \sigma(X_1 \dots X_k)}) = Y_k.$$

$$\mathbb{E}_Y(\overbrace{\mathbb{E}(X|Y)}^{g(Y)}) = \mathbb{E}(X)$$

Example 2.17. Doob Construction:

$$Y_k := \mathbb{E}(f \mid X_1 \dots X_k)$$

① $\mathbb{E}(|Y_k|) = \mathbb{E}(|\mathbb{E}(f| X_1 \dots X_k)|) \leqslant \mathbb{E}(\mathbb{E}(|f| \mid X_1 \dots X_k))$

$$= \underbrace{\mathbb{E}(|f|) < \infty}_{\text{assumption ①}}$$

② $\mathbb{E}(Y_{k+1} \mid X_1 \dots X_k) \overset{?}{=} Y_k$

$$\mathbb{E}(\mathbb{E}(f \mid X_1 \dots X_{k+1}) \mid X_1 \dots X_k) = \overbrace{\mathbb{E}(f \mid X_1 \dots X_k)}^{Y_k}$$

tower prop.

Th. 2.19

$D_k = Y_k - Y_{k-1}$ martingale

$\rightarrow \mathbb{E}\left(e^{\lambda D_k} \mid \mathcal{F}_{k-1}\right) \leq e^{\lambda^2 \nu^2 / 2} \qquad |\lambda| \leq \frac{1}{\alpha}$

$P\left(\left|\sum_{k=1}^{n} D_k\right| > t\right) \leq e^{\cdots}$